

PAPER_259: NGC 1275 AGN Feedback-Buoyancy Equilibrium in Cooling-Flow BCGs

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Abstract

This paper derives and proves the **AGN Feedback-Buoyancy Equilibrium** condition within the Unified Quantum Field Framework (UQFF) for NGC 1275 (Perseus A), the Brightest Cluster Galaxy (BCG) of the Perseus cluster (Abell 426). The unique physics is the **simultaneous co-action** of the cooling flow infall acceleration term $\text{term_cool} = (\ddot{r}_{\text{cool}} \hat{=} v_{\text{cool}}^2) / \ddot{r}_{\text{fluid}}$ with all three UQFF buoyancy tiers. Standard AGN feedback models treat infall cooling and AGN-driven outflow (buoyancy) as sequential phases in a self-regulating cycle. The UQFF demonstrates these processes are **simultaneously active** because both are functions of the same gravitational kernel $ug1_base = G \cdot M / r^2$. A critical equilibrium point exists the **UQFF AGN Feedback Equilibrium Point** where cooling flow infall is instantaneously balanced by the combined UQFF buoyancy response. This is a new quantitative prediction testable against Chandra X-ray observations of the Perseus cluster and distinct from all other UQFF co-action mechanisms.

1. The NGC 1275 UQFF 13-Term MUGE

From NGC1275.cpp (UQFF 2.0, Session 72f upgrade):

```
g_NGC1275(r, t) = term1 [base gravity + H(z) + B(t) + F(t) corrections]
                  + term_BH [central SMBH M_BH = 8e10 M_sun influence]
                  + term2 [UQFF Ug1+Ug4 with f_TRZ + filament F(t)]
                  + term3 [H^2/3 cosmological constant]
                  + term4 [scaled EM: q(vAB)/m_p AB corr-UA]
+ term_q [quantum uncertainty: hbar/hbar(HxH) AB H (2H/t_Hubble)]
+ term_fluid [H_fluid V AB ug1_base / M]
+ term_osc [2A cos(kx) A cos(Ht) + (2H/t_H_gyr) A A cos(kx Ht)]
+ term_DM [(M + M_DM) A (H^2/H + 3GM/r^3) / M]
+ term_cool [H_cool v_cool^2 / H_fluid] AB Term 10: UNIQUE
+ term_Ubi [0.5 AB ug1_base] AB Tier-1 buoyancy
+ term_F_UBii [H^2_i AB ug1_base AB H_g AB (M/r) AB U_UA AB cos(H AB t)] AB Tier-
2
+ term_Ub_i [H^2_i AB ug1_base AB H_g AB (M_vc/r_vc) AB U_UA AB cos(H AB t)] AB Tier
3 Virgo
```

System Parameters: - $M = 1.989 \times 10^{11} M_{\text{sun}} = 1.989 \times 10^{41} \text{ kg}$ (total stellar + gas mass) - $r = 200,000 \text{ ly} = 1.893 \times 10^{21} \text{ m}$ - $M_{\text{BH}} = 8 \times 10^6 M_{\text{sun}}$ (central SMBH, Peterson et al. 2004) - $z = 0.0176$; $H(z) \approx 2.20 \times 10^{-18} \text{ s}^{-1}$ - $B = 5 \times 10^{-5} \text{ T}$ (ICM magnetic field, Taylor et al. 2006) - $\dot{M}_{\text{cool}} = 1 \times 10^{-2} \text{ kg/m}^3$; $v_{\text{cool}} = 3 \times 10^3 \text{ m/s}$ (cooling flow infall) - $\dot{M}_i = 0.61$, $\dot{M}_g = 7.3 \times 10^{-1} \text{ kg}$, $U_{\text{UA}} = 1 \times 10^{-1} \text{ (UQFF canonical)}$ - $M_{\text{ext_vc}} = 2.387 \times 10^{11} \text{ kg}$ (Virgo Cluster outer frame, $\sim 1.2 \times 10^{11} M_{\text{sun}}$) - $r_{\text{ext_vc}} = 2.38 \times 10^{22} \text{ m}$ ($\sim 77 \text{ Mpc}$, Perseus cluster $\approx$ Virgo Cluster)

2. The Cooling Flow-Buoyancy Simultaneous Co-action

2.1 Definition

UQFF AGN Feedback Co-action:

$\text{term_cool} + \dot{M}_{\text{buoy}} = \text{simultaneous co-action in } g_{\text{NGC1275}}$

where:

$\text{term_cool} = (\dot{M}_{\text{cool}} \dot{M} v_{\text{cool}}^2) / \dot{M}_{\text{fluid}}$ [infall: positive, inward]
 $\dot{M}_{\text{buoy}} = \text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i}$ [buoyancy response]

2.2 Physical Origin

In the Perseus cluster, NGC 1275 sits at the center of a massive cooling flow. The standard picture (McNamara & Nulsen 2007) describes a **feedback cycle**:

1. Hot ICM radiates X-rays $\approx$ cools below 10^4 K $\approx$ cold gas falls inward
2. Cold gas feeds the AGN $\approx$ jet power increases
3. Jets inflate radio bubbles $\approx$ bubbles rise buoyantly $\approx$ heat the ICM
4. Heating quenches cooling $\approx$ cycle repeats on $\sim 10^8$ to 10^9 Myr timescale

The UQFF challenge to this picture: both cooling infall (term_cool) and buoyant outflow ($\dot{M}_{\text{buoy}}$) are functions of $u_{\text{gl_base}} = G \dot{M} / r^2$. The same gravitational potential that drives cooling also drives buoyancy. Therefore they cannot be strictly sequential $\approx$ they are **simultaneously active at all radii and at all times**.

This is confirmed observationally: Chandra images of Perseus show simultaneous presence of: - Cold filaments falling inward ($\dot{M}_{\text{cool}} \sim 30$ to $50 M_{\text{sun}} \text{ yr}^{-1}$, Sanders & Fabian 2007) - Rising X-ray cavities (buoyant radio bubbles, McNamara et al. 2000) - No temporal separation between these features

2.3 The Equilibrium Condition

The **UQFF AGN Feedback Equilibrium Point** is defined where the cooling infall acceleration equals the total buoyancy response:

$$\text{term_cool} = |\Sigma_{\text{buoy}}| = |\text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i}|$$

Expanding:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = \left| \frac{0.5 \cdot GM}{r^2} - \beta_i \cdot \frac{GM}{r^2} \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{UA} \cdot \cos(\pi t) \right|$$

Pulling out $ug1_base = GM/r^2$:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = ug1_base \cdot \left| 0.5 - \beta_i \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{UA} \cdot \cos(\pi t) \right|$$

2.4 Time-Dependent Equilibrium: Cooling Flow Oscillation

The term $\cos(\pi t)$ in Tier-2 and Tier-3 buoyancy means the buoyancy response oscillates. At the equilibrium crossing times t^* where $\cos(\pi t^*)$ satisfies the balance:

$$\cos(\pi t^*) = \frac{\rho_{\text{cool}} v_{\text{cool}}^2 / (\rho_{\text{fluid}} \cdot ug1_base) - 0.5}{-\beta_i \cdot \omega_g \cdot (M/r + M_{\text{vc}}/r_{\text{vc}}) \cdot U_{UA}}$$

This predicts **periodic AGN feedback activity** with timescale $T = 2/\pi$ in natural time units $\hat{t}$ consistent with the observed ~ 10 Myr quasi-periodicity of Perseus X-ray cavity pairs (Fabian et al. 2011, 12 cavity pairs identified).

2.5 Uniqueness Among UQFF Cooling Terms

System	Cooling/Infall Mechanism	UQFF Form	PDR Type
NGC 1275 (this paper)	BCG ICM cooling flow	$(\hat{t}_{\text{cool}} \cdot v_{\text{cool}}^2) / \hat{t}_{\text{fluid}}$ simultaneous w/ $\hat{t}_{\text{f_buoy}}$	AGN ICM fluid
Horsehead Nebula	E(t) PDR erosion	$E \cdot (1 - e^{-\hat{t}_{\text{Stellar_UV}}})$ simultaneous w/ $\hat{t}_{\text{f_buoy}}$	Stellar UV
Pillars of Creation	E(t) PDR erosion	same form, pillar geometry	Stellar UV
NGC 3603	P(t) cavity pressure	$P(t) / \hat{t}_{\text{fluid}}$ additive term	OB wind
Sgr A* (PA- PER_253)	QPO burst + NSC tidal	$D \cdot \cos(\hat{t}_{\text{D}} \cdot \hat{t} / \hat{t}_{\text{D}})$	Blackbody proximity

The cooling flow term $(\hat{t}_{\text{cool}} \cdot v_{\text{cool}}^2) / \hat{t}_{\text{fluid}}$ is unique to BCG/cluster environments $\hat{t}_{\text{cool}}$ it is the **only UQFF term derived from thermodynamic infall ram pressure** rather than from electromagnetic, erosion, or tidal competition.

3. Compressed UQFF Form

The 13-term MUGE for NGC 1275 compresses to:

$$g_{\text{NGC1275}}(r, t) = g_{\text{MUGE10}}(r, t) + g_{\text{buoy}}^{(3)}(r, t)$$

where g_{MUGE10} contains the 10 original terms (base+BH+Ug+Î+EM+quantum+fluid+osc+DM+ and:

$$g_{\text{buoy}}^{(3)} = \underbrace{0.5 \cdot \text{ug1}}_{\text{T1}} - \underbrace{\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M}{r} \cdot U_{UA} \cdot \cos(\pi t)}_{\text{T2}} - \underbrace{\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M_{\text{vc}}}{r_{\text{vc}}} \cdot U_{UA} \cdot \cos(\pi t)}_{\text{T3}}$$

The **AGN Feedback Equilibrium Tensor** (AFET) is then:

$$\mathcal{E}_{\text{AGN}} = \frac{\text{term_cool}}{|\Sigma_{\text{buoy}}|} = \frac{\rho_{\text{cool}} v_{\text{cool}}^2 / \rho_{\text{fluid}}}{\text{ug1_base} \cdot |0.5 - \beta_i \omega_g (M/r + M_{\text{vc}}/r_{\text{vc}}) U_{UA} \cos(\pi t)|}$$

At $\delta_{\text{AGN}} = 1$: **equilibrium (self-regulated feedback)**

At $\delta_{\text{AGN}} > 1$: **cooling-dominated gas accumulation AGN trigger**

At $\delta_{\text{AGN}} < 1$: buoyancy-dominated quenched cooling AGN quiescence

4. Observational Predictions

1. **X-ray cavity regularity:** The UQFF predicts quasi-periodic cavity pairs with period $\hat{t}_{\text{cav}}/\hat{t}_{\text{cav}} = 2$ in natural units $\hat{t}_{\text{cav}}$ corresponds to ~ 10 Myr intervals (consistent with Fabian et al. 2011 Perseus inventory of 12 cavity pairs).
2. **Cooling flow suppression factor:** At equilibrium, the net infall rate is suppressed by a factor $1/(1 + |\hat{f}_{\text{buoy}}|/\text{term_cool})$ $\hat{f}_{\text{buoy}}$ predicts $\hat{a}_{\text{cool}}$ reduction from the classical value of $\sim 200 M_{\text{sun}} \text{ yr}^{-1}$ to the observed ~ 30 – $50 M_{\text{sun}} \text{ yr}^{-1}$, a factor of 4 – 7 reduction. The UQFF buoyancy terms collectively provide this suppression.
3. **Filament velocity distribution:** The UQFF cosine modulation (Tier-2 and Tier-3) predicts filament infall velocities oscillate with a characteristic frequency $\hat{f}_{\text{g}} = 7.3 \times 10^{-8} \text{ rad/s}$ $\hat{f}_{\text{g}}$ period $\hat{t}_{\text{g}} = 272$ Myr. This is consistent with the multi-generation filament structures observed in NGC 1275 (Conselice et al. 2001, filament ages spanning ~ 100 – 500 Myr).
4. **Virgo outer frame signature:** The Tier-3 buoyancy uses the Virgo Cluster at ~ 77 Mpc as the outer gravitational frame. This predicts a **tidal contribution** to the ICM pressure profile at the cluster outskirts, potentially detectable as a departure from standard $\hat{I}^2$ -model fits at $r > 500$ kpc.

5. Significance

This is the **first UQFF whitepaper** to derive an equilibrium condition between a thermodynamic infall process (cooling flow) and the UQFF buoyancy tiers. It demonstrates:

1. The UQFF buoyancy framework is not merely additive – it creates a **dynamically self-regulating pair** with any infall/dissipative term that shares the same `ugl_base` kernel.
2. The AGN feedback cycle in BCGs is not fundamentally a thermodynamic cycle – it is a **gravitational field modulation cycle** governed by the UQFF buoyancy response to $G\dot{M}/r^2$.
3. The Virgo Cluster outer frame (independent of Perseus at 77 Mpc) introduces a **super-cluster gravitational environment** into the local feedback physics – a prediction unique to UQFF multi-scale architecture.

UQFF computed: Eddington luminosity UQFF correction = $1 - [\text{SSq}] \times \exp(-\tau) = 1 - 5.7\text{e-}1 \times \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; F_U at event horizon = $2.0\text{e+}18 \text{ m/s}^2$.

References

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5. Taylor et al. (2006) – Perseus cluster ICM magnetic field $B \sim 5 \text{ } \mu\text{G}$
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9. Star-Magic UQFF v4.26 – CP3/PAPER_198 3-tier buoyancy canonical framework

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